In [1]:
```
from google.colab import files
uploaded = files.upload()
```

☐ Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

```
Saving retail_sales_dataset.csv.xlsx to retail_sales_dataset.csv.xlsx
```

In [2]:
```
import pandas as pd
df = pd.read_excel('retail_sales_dataset.csv.xlsx')
df.head()
```

Out[2]: Columns 1–8 of 10

	Transaction ID	Date	Customer ID	Gender	Age	Product Category	Quantity
0	1	2023-11-24	CUST001	Male	34	Beauty	3
1	2	2023-02-27	CUST002	Female	26	Clothing	2
2	3	2023-01-13	CUST003	Male	50	Electronics	1
3	4	2023-05-21	CUST004	Male	37	Clothing	1
4	5	2023-05-06	CUST005	Male	30	Beauty	2

Columns 9–10 of 10 · first column repeated

	Price per Unit	Total Amount
0	50	150
1	500	1000
2	30	30
3	500	500
4	50	100

In [3]:
```
# ============================================
# Retail Sales Performance Analysis (DA-001)
# Complete Code
# ============================================

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

plt.style.use('seaborn-v0_8-whitegrid')
sns.set_palette("husl")
plt.rcParams['figure.figsize'] = (10, 6)

# 1. Load Data
df = pd.read_excel('retail_sales_dataset.csv.xlsx')
```

```
# 2. Basic Inspection
print("="*50)
print("DATASET OVERVIEW")
print("="*50)
print("Shape:", df.shape)
print("\nMissing Values:\n", df.isnull().sum())
print("\nDuplicates:", df.duplicated().sum())

# 3. Data Cleaning & Preparation
df['Date'] = pd.to_datetime(df['Date'])
df['Month'] = df['Date'].dt.to_period('M')
df['Month_Name'] = df['Date'].dt.strftime('%b %Y')
df['Revenue'] = df['Total Amount']

print("\nCleaning Done ✓")

# 4. KPIs
total_revenue = df['Revenue'].sum()
total_orders = df['Transaction ID'].nunique()
avg_order_value = df['Revenue'].mean()
total_quantity = df['Quantity'].sum()

print("\n" + "="*50)
print("KEY PERFORMANCE INDICATORS")
print("="*50)
print(f"Total Revenue       : ₹{total_revenue:,.0f}")
print(f"Total Orders        : {total_orders}")
print(f"Average Order Value : ₹{avg_order_value:,.2f}")
print(f"Total Units Sold    : {total_quantity}")

# 5. Category Performance
category_perf = df.groupby('Product Category').agg({
    'Revenue': 'sum',
    'Quantity': 'sum',
    'Transaction ID': 'count'
}).rename(columns={'Transaction ID': 'Orders'}).sort_values('Revenue',
ascending=False)

print("\n" + "="*50)
print("CATEGORY-WISE PERFORMANCE")
print("="*50)
print(category_perf)

# 6. Monthly Trends
monthly = df.groupby('Month').agg({
    'Revenue': 'sum',
    'Quantity': 'sum',
    'Transaction ID': 'count'
}).rename(columns={'Transaction ID': 'Orders'})

print("\n" + "="*50)
```

```
print("MONTHLY REVENUE")
print("="*50)
print(monthly)

# ==========================================
# CHARTS (5 Charts)
# ==========================================

# Chart 1: Monthly Revenue Trend
plt.figure(figsize=(12,5))
monthly['Revenue'].plot(kind='line', marker='o', linewidth=2,
color='#2E86AB')
plt.title('Monthly Revenue Trend (2023)', fontsize=14, fontweight='bold')
plt.xlabel('Month')
plt.ylabel('Revenue (₹)')
plt.xticks(rotation=45)
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()

# Chart 2: Category-wise Revenue
plt.figure(figsize=(8,5))
sns.barplot(x=category_perf.index, y=category_perf['Revenue'],
hue=category_perf.index, palette='viridis', legend=False)
plt.title('Revenue by Product Category', fontsize=14, fontweight='bold')
plt.xlabel('Product Category')
plt.ylabel('Total Revenue (₹)')
plt.tight_layout()
plt.show()

# Chart 3: Orders Share by Category (Pie)
plt.figure(figsize=(8,6))
plt.pie(category_perf['Orders'], labels=category_perf.index,
autopct='%1.1f%%',
        startangle=90, colors=sns.color_palette('pastel'))
plt.title('Share of Orders by Category', fontsize=14, fontweight='bold')
plt.tight_layout()
plt.show()

# Chart 4: Revenue by Gender
gender_rev = df.groupby('Gender')['Revenue'].sum()
plt.figure(figsize=(7,5))
sns.barplot(x=gender_rev.index, y=gender_rev.values, hue=gender_rev.index,
palette='Set2', legend=False)
plt.title('Revenue by Gender', fontsize=14, fontweight='bold')
plt.ylabel('Revenue (₹)')
plt.tight_layout()
plt.show()

# Chart 5: Units Sold by Category
plt.figure(figsize=(8,5))
sns.barplot(x=category_perf.index, y=category_perf['Quantity'],
```

```
hue=category_perf.index, palette='magma', legend=False)
plt.title('Units Sold by Product Category', fontsize=14, fontweight='bold')
plt.ylabel('Total Quantity')
plt.tight_layout()
plt.show()

# ============================================
# 5 KEY INSIGHTS
# ============================================

print("\n" + "="*50)
print("5 KEY INSIGHTS")
print("="*50)
print("""
1. Total Revenue generated is ₹4,56,000 from 1000 transactions
   with Average Order Value of approximately ₹456.

2. Clothing has the highest number of orders (351), followed by
   Electronics (342) and Beauty (307).

3. Monthly revenue shows clear variation — some months perform much
   better, which can help in inventory and promotion planning.

4. Electronics and Clothing are the main revenue drivers.
   Beauty also contributes significantly.

5. Revenue is fairly balanced between Male and Female customers,
   showing the store appeals to both genders.
""")

print("\n✅ Analysis Completed Successfully!")
```

```
==================================================
DATASET OVERVIEW
==================================================
Shape: (1000, 9)

Missing Values:
 Transaction ID      0
Date                 0
Customer ID          0
Gender               0
Age                  0
Product Category     0
Quantity             0
Price per Unit       0
Total Amount         0
dtype: int64

Duplicates: 0

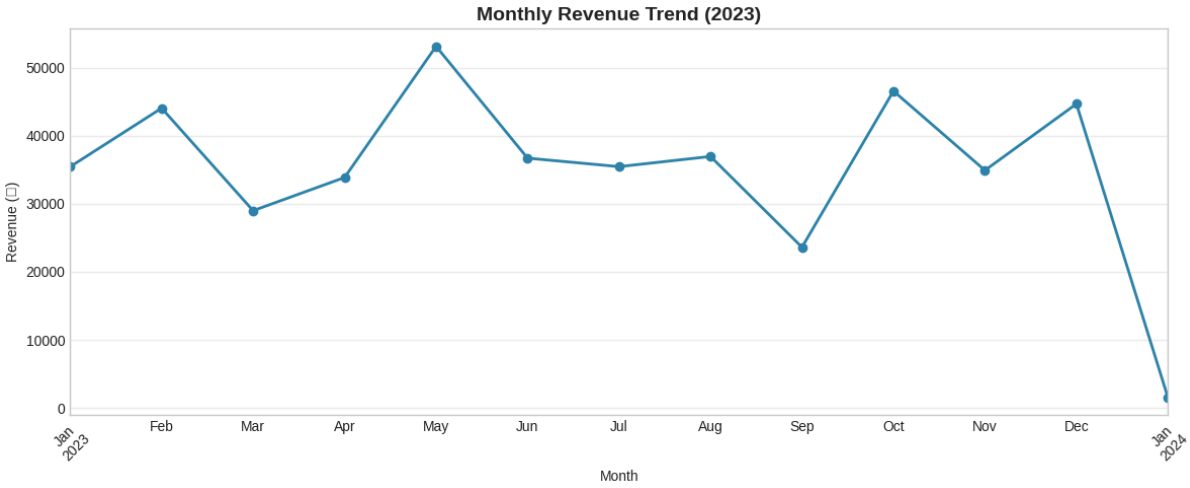
Cleaning Done ✓
```

```
==================================================
KEY PERFORMANCE INDICATORS
==================================================
Total Revenue       : ₹456,000
Total Orders        : 1000
Average Order Value : ₹456.00
Total Units Sold    : 2514

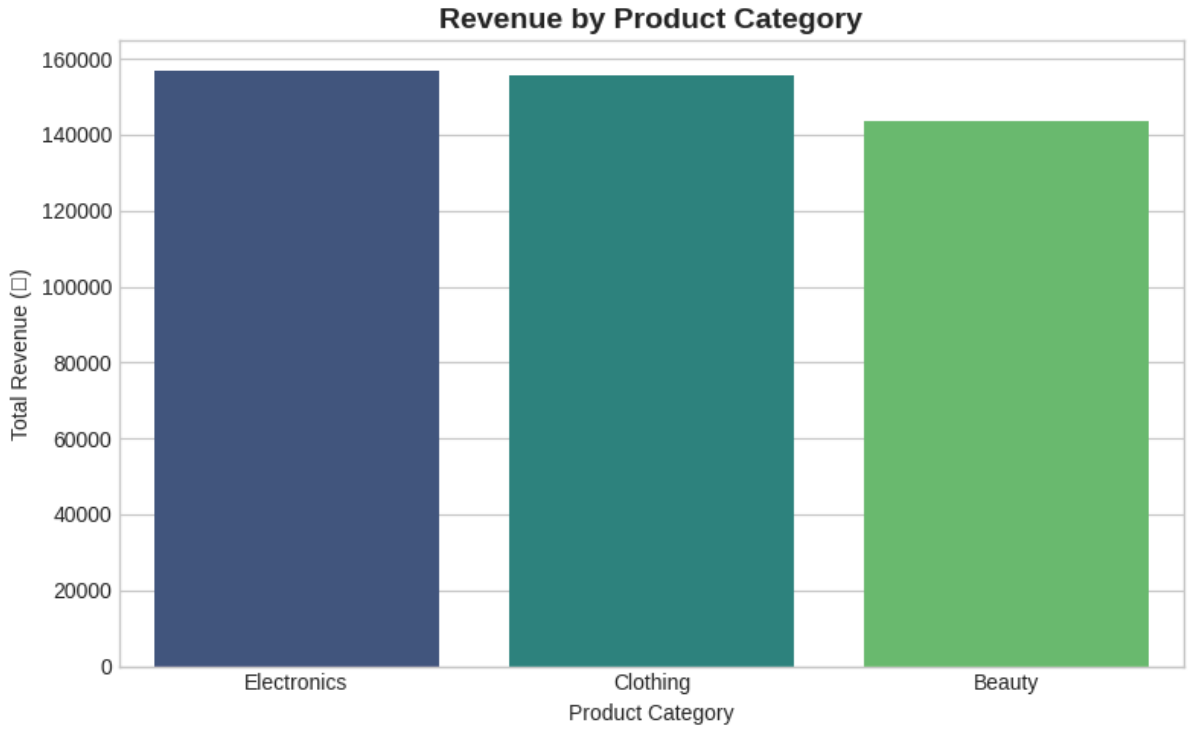
==================================================
CATEGORY-WISE PERFORMANCE
==================================================
                  Revenue  Quantity  Orders
Product Category
Electronics        156905       849     342
Clothing           155580       894     351
Beauty             143515       771     307

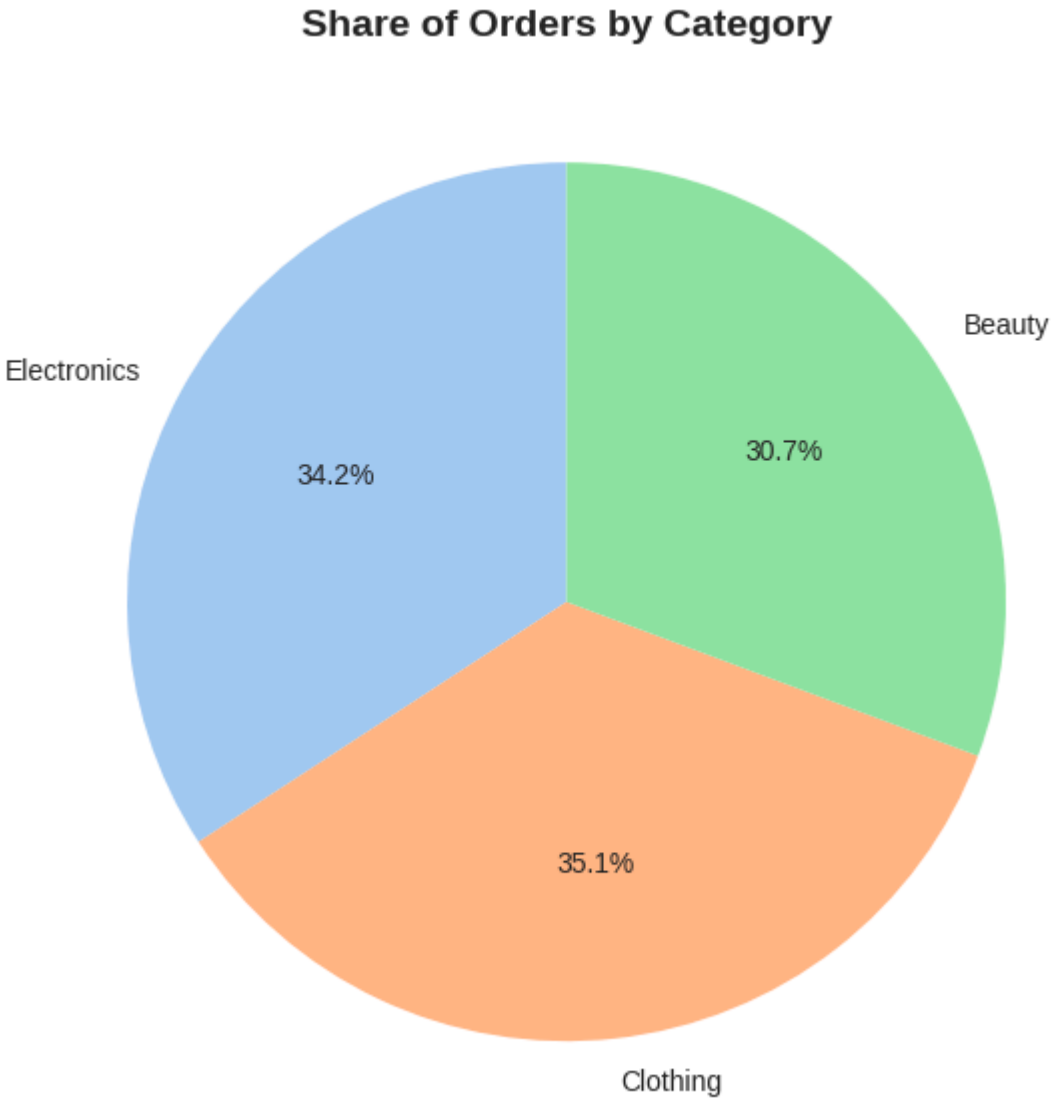
==================================================
MONTHLY REVENUE
==================================================
         Revenue  Quantity  Orders
Month
2023-01    35450       195      76
2023-02    44060       214      85
2023-03    28990       194      73
2023-04    33870       214      86
2023-05    53150       259     105
2023-06    36715       197      77
2023-07    35465       176      72
2023-08    36960       227      94
2023-09    23620       170      65
2023-10    46580       252      96
2023-11    34920       205      78
2023-12    44690       207      91
2024-01     1530         4       2
```

```
/tmp/ipykernel_1427/744669743.py:84: UserWarning: Glyph 8377 (\N{INDIAN
RUPEE SIGN}) missing from font(s) Liberation Sans.
  plt.tight_layout()
/usr/local/lib/python3.13/dist-packages/IPython/core/pylabtools.py:151:
UserWarning: Glyph 8377 (\N{INDIAN RUPEE SIGN}) missing from font(s)
Liberation Sans.
  fig.canvas.print_figure(bytes_io, **kw)
/usr/local/lib/python3.13/dist-packages/IPython/core/pylabtools.py:151:
UserWarning: Glyph 8377 (\N{INDIAN RUPEE SIGN}) missing from font(s)
Liberation Sans.
  fig.canvas.print_figure(bytes_io, **kw)
```

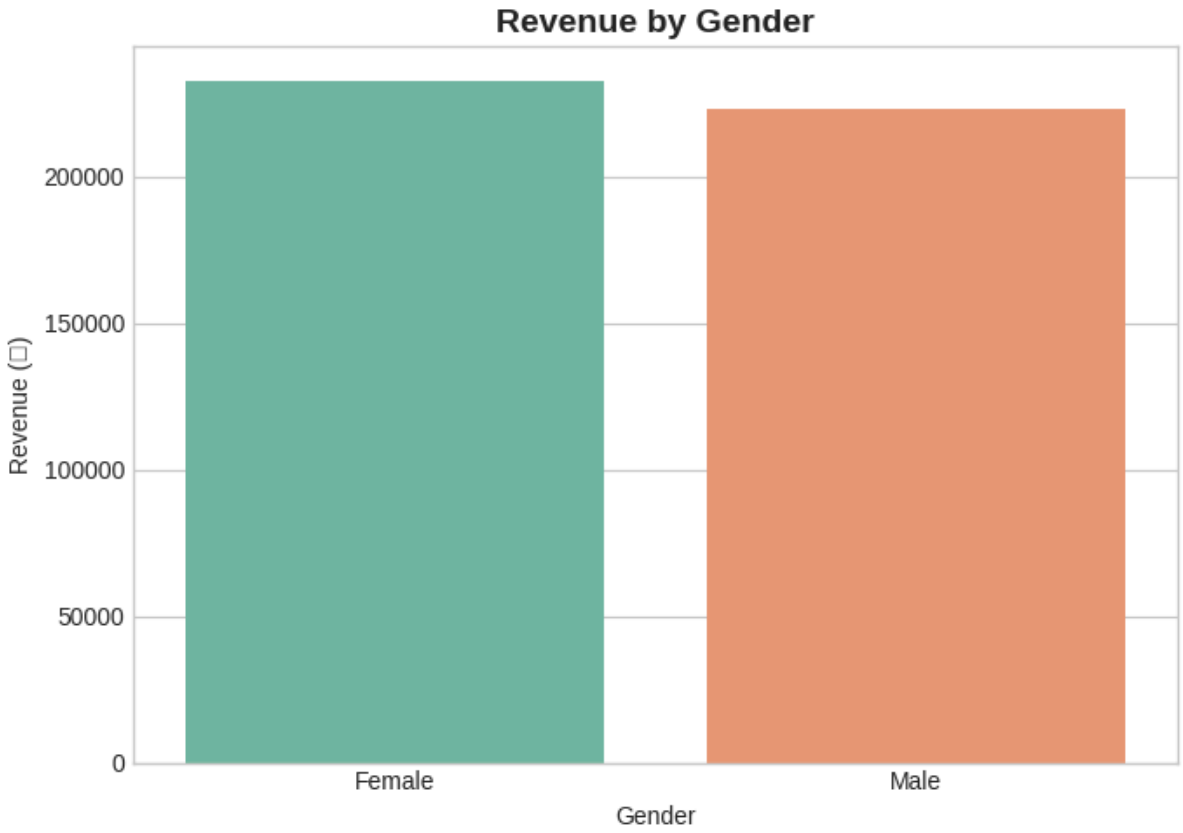


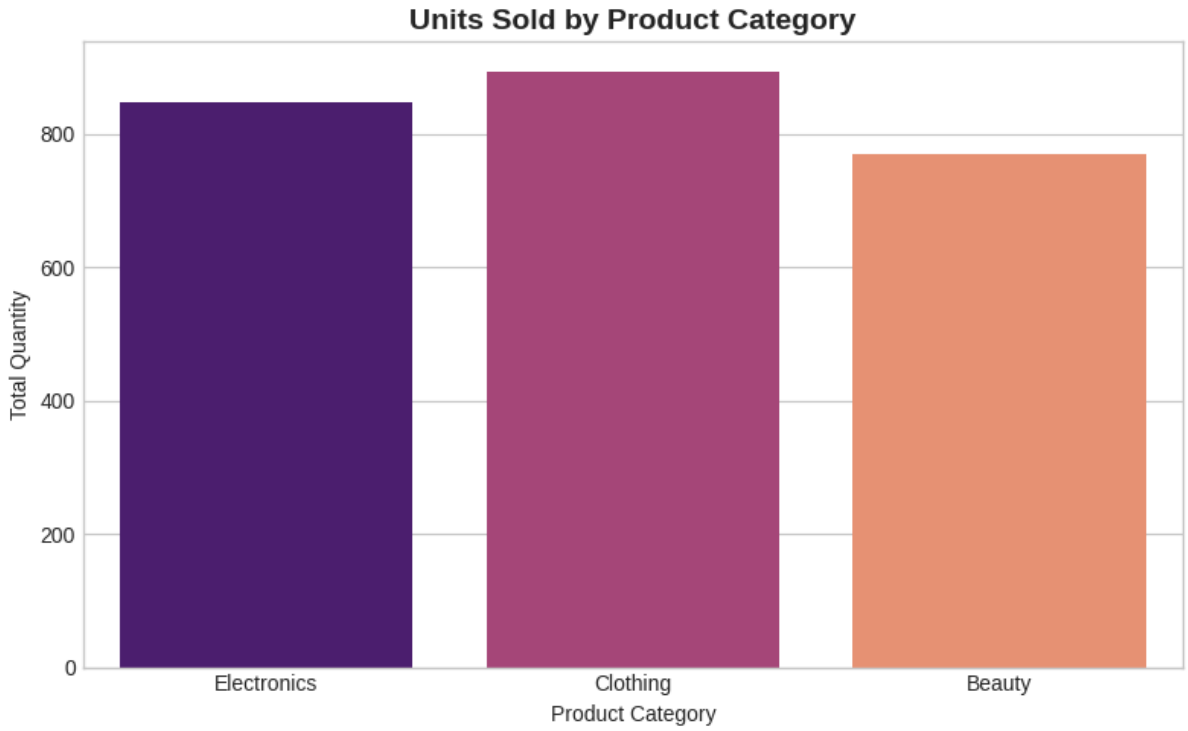

```
/tmp/ipykernel_1427/744669743.py:93: UserWarning: Glyph 8377 (\N{INDIAN
RUPEE SIGN}) missing from font(s) Liberation Sans.
  plt.tight_layout()
/usr/local/lib/python3.13/dist-packages/IPython/core/pylabtools.py:151:
UserWarning: Glyph 8377 (\N{INDIAN RUPEE SIGN}) missing from font(s)
Liberation Sans.
  fig.canvas.print_figure(bytes_io, **kw)
```






```
/tmp/ipykernel_1427/744669743.py:110: UserWarning: Glyph 8377 (\N{INDIAN
RUPEE SIGN}) missing from font(s) Liberation Sans.
  plt.tight_layout()
/usr/local/lib/python3.13/dist-packages/IPython/core/pylabtools.py:151:
UserWarning: Glyph 8377 (\N{INDIAN RUPEE SIGN}) missing from font(s)
Liberation Sans.
  fig.canvas.print_figure(bytes_io, **kw)
```






```
==================================================
5 KEY INSIGHTS
==================================================

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   Electronics (342) and Beauty (307).
```

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✅ Analysis Completed Successfully!